

RegRadar

A multi-agent regulatory intelligence system that monitors, summarizes, and triages regulatory filings (SEC, FDA, FINRA) in real time.

Portfolio Tier: High	Timeline: 8–12 weeks	Difficulty: Senior
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REAL-WORLD PROBLEM IT SOLVES

Compliance teams at mid-market companies spend 20–40 hrs/week manually reading regulatory filings. A new SEC rule or FDA guidance can trigger legal review, product changes, and board updates — but most orgs only catch it days late via Google Alerts. RegRadar ingests filings continuously, routes them through a triage agent, extracts structured risk signals, and delivers a Slack/email brief to the right person within minutes of publication.

HUGGING FACE TASKS USED

Summarization	Document Question Answering	Text Classification	Zero-Shot Classification
Feature Extraction	Table Question Answering		

Summarization condenses 80-page filings into 3-sentence briefs. Zero-shot classification routes documents to the right compliance domain without labeled training data. Feature extraction powers the RAG retrieval layer. Table QA handles structured exhibits and financial tables inside PDFs.

RECOMMENDED TECH STACK

Orchestration	LangGraph (multi-agent DAG)
RAG	LlamaIndex + pgvector
Models	HF Inference API + GPT-4o
Eval	Ragas + LangSmith
Ingestion	SEC EDGAR API, FDA RSS, Prefect

Serving	FastAPI + Redis queue
Infra	Docker + Fly.io / Railway
Storage	Supabase (Postgres + pgvector)
Alerts	Slack Webhooks, SendGrid
Observability	Langfuse or Helicone

MAPS TO AI ENGINEERING JD REQUIREMENTS

- ✓ Multi-agent orchestration with LangGraph DAGs and inter-agent state sharing
- ✓ RAG pipeline design: chunking strategy, hybrid search, re-ranking, context window management
- ✓ LLM evaluation: Ragas metrics (faithfulness, context recall), ROUGE-L on summaries, human-in-the-loop feedback
- ✓ LLMops: prompt versioning, cost monitoring, model A/B testing, drift detection
- ✓ Production API: rate-limited FastAPI, async task queues, webhook delivery
- ✓ Real data sources: live SEC EDGAR, FDA Drugs@FDA, FINRA rulemaking feeds
- ✓ Domain-specific model progression: zero-shot to few-shot to adapter fine-tune

RESUME-READY IMPACT METRICS

<3 min	6 agents	0.87+
Filing to Slack alert latency	Orchestrated via LangGraph	Ragas faithfulness score
500K+	4 eval	<\$0.03
Real regulatory docs indexed	Automated metrics tracked	Cost per document brief

WHAT MAKES THIS STAND OUT VS A BASIC DEMO

Eval harness is first-class

Every prompt change runs a regression suite before it ships. Ragas scores are logged to LangSmith and graphed over time.

Thoughtful chunking strategy

Regulatory PDFs have tables, footnotes, and cross-references. The project documents chunking decisions and their measurable effect on retrieval quality.

Cost discipline

A tiered routing layer sends low-priority docs through a cheaper HF-hosted model and high-priority ones through GPT-4o. Cost per document is tracked.

Failure modes are documented

Hallucination rate per document type, recall gaps on tabular data, latency spikes under load. A known-limitations section signals engineering maturity.